Machine Learning Mini Project

Prediction of YouTube Video Reach Levels in HK & TW Markets

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Project Objectives

The primary objective of this project is to develop and evaluate a predictive model to estimate a YouTube video's "Reach Level" within the Hong Kong and Taiwan markets prior to its publication, while identifying the key drivers that significantly influence video performance.

Dataset Description

The dataset (Appendix A)from Kaggle about trending YouTube videos globally and provided a comprehensive perspective, containing over 7.63 million video records spanning 110 countries and regions. We filtered the data by the ‘video_trending_country' column, isolating Hong Kong and Taiwan. This resulted in a refined subset of 24,086 records, providing a targeted foundation for training models.

The dataset consists of 28 columns, which can be categorized into two dimensions: Video-related features and Channel-related features.Video-related features include details about the video itself, such as the category, publishing time, video duration, and performance metrics like views, likes, and comments. Channel-related features are focused on the creator's profile, such as the total subscriber count, total channel views, and the total number of videos already published by the channel. The dataset reveals a highly skewed distribution of "Reach Level", Level 1 (Low Reach) accounts for ~82% of the data, while Level 2 (~10%) and Level 3 (~8%) are minority classes, posing a significant challenge for predictive accuracy.

We converted time-related columns, such as video_published_at and channel_published_at, into the datetime64 format. This allowed for precise time-based calculations and feature extraction in later stages. Regarding missing values, the channel_country column was filled with the label ’Unknown’, while missing entries in video_tags were filled with ‘None’, representing a lack of information for those records. To make the data readable we manually mapped the video_trending_country column into integers: Hong Kong was mapped to 0 and Taiwan to 1.

Methodology

The target variable for this project is “Reach Level”.We split video views into three levels using these ranges:

Level 1: Less than 278,219 views.
Level 2: 278,219 to 933,977 views.
Level 3: 933,978 views or more.

At the beginning, a basic set of 11 features was established. However, through iterative testing and hyperparameter tuning, the results remained unsatisfactory, primarily due to the low recall and Macro F1-score for Level 2 videos. The model struggled to distinguish mid-tier performance from the majority class (Level 1).

To address this, we refined our feature set by excluding several columns with no predictive value, such as video_definition. More importantly, we introduced **five powerful new features** to provide the model with deeper insights. A key improvement was the addition of duration_seconds. While the original dataset included an is_shorts column, it was too simplistic to represent the actual length of a video. By including the exact duration in seconds, the model could learn the impact of video length more precisely.

Furthermore, we integrated **channel-related features** (subscribers, total views) and performed transformations such as calculating channel_age_days to represent creator seniority. **The final processed dataset (Appendix B)** consists of **37 columns**, with the training set narrowed down to **14 key features**:

Channel Authority: subscribers, channel_views, video_count, channel_age, country_int.
Video Characteristics: is_shorts, duration_seconds, category_id, trending_country, licensed_content.
Publishing Strategy: title_length, tag_count, publish_hour, day_of_week.

<u>Model selection</u>
We compared two ensemble models to evaluate their performance:

Random Forest (Baseline): A reliable model that handles non-linear relationships well, used to establish a performance baseline.

XGBoost (Optimized): An advanced gradient boosting algorithm selected for its superior handling of **imbalanced data**. By utilizing **sample weighting**, XGBoost was prioritized to focus on minority classes (Level 2 and 3).

Hyperparameter Tuning

Our strategy focused on **Learning Rate, Max Depth,** and **N-estimators**. To specifically address class imbalance, we applied:

Sample Weighting: Utilized class_weight='balanced' to prioritize minority classes.
min_child_weight (0.1): Allowed the model to learn detailed patterns from smaller, specific data groups.
Regularization: Implemented **Early Stopping (50 rounds)** and monitored **Learning Curves** (using both X_train and X_val) to prevent overfitting.

Analysis

XGBoost

The final XGBoost model (Appendix C)achieved an Accuracy of 0.88 and a Macro F1-score of 0.74. Most importantly, the Recall for Level 2 reached 0.63, representing a significant improvement over our baseline attempts and demonstrating the model's enhanced ability to capture minority classes.

To identify the performance ceiling, we compared **an aggressive configuration (Appendix D)(LR 0.02, Depth** 10) against our final conservative set (LR 0.03, Depth 8). The analysis of the **Learning Curves (Appendix E)** revealed:

The Overfitting Signal: In the aggressive set, validation loss reached a minimum of 0.30582 at the 600th iteration but **rebounded to 0.30629** by the 640th iteration. This indicated that the model began to "memorize" training noise rather than learning generalizable trends.

Recall Decay: Consequently, Level 2 Recall **dropped from 0.63 to 0.60**, proving that excessive complexity reduced predictive accuracy for mid-tier videos.

Optimal Convergence: Our final model reached an optimal stopping point at 721 iterations **(Loss: 0.30630)** with a stable, non-rebounding curve, ensuring superior robustness for new data.

Visualization of results (Appendix E):

Confusion Matrix: Confirms high precision for viral videos (Level 3), with most errors occurring only between adjacent levels (L1/L2), proving logical consistency.

Feature Importance (Gain): Identifies channel_subscriber_count and duration_seconds as the top drivers of reach. This validates our hypothesis that combining channel authority with precise video length is crucial for prediction.

F1-Score Comparison: A bar chart highlighting the model's balanced predictive power across all three levels, effectively overcoming the class imbalance challenge.

Random Forest

The Random Forest model (Appendix F) served as our robust baseline, achieving an **Accuracy of 0.90** and a **Weighted F1-score of 0.89**. While it shared a similar Macro F1-score (0.74) with XGBoost, it demonstrated exceptional stability in predicting the majority class (Level 0), with a **Recall of 0.97**. For the high-reach viral category (Level 2), it achieved a strong **Precision of 0.80** and an **F1-score of 0.74**, proving its reliability in identifying top-performing content.

Learning Curve Diagnostics (Appendix F)

Training Score (1.0): The horizontal line at 1.0 reflects the model's high learning capacity and ability to perfectly fit the training data.

Validation Score (0.86 to 0.90): The steady ascent of the validation curve confirms that as the training size increased, the model successfully improved its generalization, reaching a stable plateau at 0.90.

Visualization of Results (Appendix G&H)

Confusion Matrix: Highlights the model's strength in identifying Level 0 videos. The primary challenge remains the "Recall Gap" in Level 1 (0.45), where mid-tier videos are often conservatively classified as Level 0.

Feature Importance (MDI): Identified **channel_subscriber_count** as the primary driver (21%), supporting the "Winner-Takes-All" market theory.

Challenges and Limitations

Severe Class Imbalance: The dataset was heavily skewed toward Level 1 (Normal videos), making it difficult for models to identify the "rare event" of going trending (Level 3).

Lack of Non-Structured Data: Our model relies solely on metadata. It cannot analyze the actual **video content, thumbnail visual appeal, or editing quality**, which are often the true drivers of virality.

Data Latency: The dataset provides a snapshot in time. It does not account for real-time external events (e.g., breaking news or memes) that trigger sudden spikes in views.

Future Work

NLP Sentiment Analysis: Analyzing the top 100 comments of each video to see if "controversy" or "high engagement" correlates with Level 3 status.
Deep Learning Models: Implementing Recurrent Neural Networks (RNNs) or LSTMs to capture the sequential nature of trending dates.

Conclusion

Random Forest: Provided a robust baseline with an **overall accuracy of 90%** and the highest **F1-Score (0.74)** for Level 3 (Trending) videos.
XGBoost: While slightly lower in accuracy (88%), it proved superior in identifying complex, non-linear drivers of virality.

Identification of Key Drivers
Through the Feature Importance analysis provided by **XGBoost**, we successfully identified the top three drivers that significantly influence video performance in the regional market:

Subscriber Count (Authority): Both models identified this as the strongest predictor. This confirms a **"winner-takes-all"** dynamic, where a large established fan base acts as a primary safety net and a springboard for trending.
Video Category (Niche Selection): Content "Type" has a decisive impact on reach. Our data shows that certain categories (e.g., Entertainment or Music) inherently possess higher viral potential in the HK/TW markets, suggesting that **choosing the right niche** is more critical than high-frequency posting.

Country (Regional Targeting): The model identified "Country" as a significant variable, highlighting distinct audience preferences and algorithmic variations between Hong Kong and Taiwan. Tailoring content to specific regional trends is essential for maximizing reach.

3. Final Insights & Strategic Takeaways

Our research demonstrates that predicting YouTube success is highly feasible. While **long-term subscriber accumulation** remains the most stable path to success, creators can maximize their odds through **precise category positioning** and **localized regional strategies.**

Reference

https://www.kaggle.com/datasets/canerkonuk/youtube-trending-videos-global

Distribution of Work

PPT: Ian and Larry

Coding: Ian and Larry

Appendix

A. Original Dateset

```
 #   Column                        Non-Null Count   Dtype
---  ------                        --------------   -----
 0   video_id                      140456 non-null  object
 1   video_published_at            140456 non-null  object
 2   video_trending__date          140456 non-null  object
 3   video_trending_country        140456 non-null  object
 4   channel_id                    140456 non-null  object
 5   video_title                   140456 non-null  object
 6   video_description             132482 non-null  object
 7   video_default_thumbnail       140452 non-null  object
 8   video_category_id             140212 non-null  object
 9   video_tags                    104789 non-null  object
 10  video_duration                140452 non-null  object
 11  video_dimension               140452 non-null  object
 12  video_definition              140452 non-null  object
 13  video_licensed_content        140452 non-null  object
 14  video_view_count              140449 non-null  float64
 15  video_like_count              139040 non-null  float64
 16  video_comment_count           139962 non-null  float64
 17  channel_title                 140452 non-null  object
 18  channel_description           127445 non-null  object
 19  channel_custom_url            133575 non-null  object
 20  channel_published_at          140452 non-null  object
 21  channel_country               118377 non-null  object
 22  channel_view_count            140420 non-null  float64
 23  channel_subscriber_count      140452 non-null  float64
 24  channel_have_hidden_subscribers 140452 non-null  object
 25  channel_video_count           140452 non-null  float64
 26  channel_localized_title       140452 non-null  object
 27  channel_localized_description 127445 non-null  object
dtypes: float64(6), object(22)
memory usage: 30.0+ MB
```

B. The final processed dataset

```
 0   video_id                      24086 non-null  object
 1   video_published_at            24086 non-null  datetime64[ns, UTC]
 2   video_trending__date          24086 non-null  object
 3   video_trending_country        24086 non-null  int64
 4   channel_id                    24086 non-null  object
 5   video_title                   24086 non-null  object
 6   video_description             24086 non-null  object
 7   video_default_thumbnail       24086 non-null  object
 8   video_category_id             24086 non-null  int64
 9   video_tags                    24086 non-null  object
 10  video_duration                24086 non-null  object
 11  video_dimension               24086 non-null  object
 12  video_definition              24086 non-null  int64
 13  video_licensed_content        24086 non-null  int64
 14  video_view_count              24086 non-null  float64
 15  video_like_count              24086 non-null  float64
 16  video_comment_count           24086 non-null  float64
 17  channel_title                 24086 non-null  object
 18  channel_description           24086 non-null  object
 19  channel_custom_url            24086 non-null  object
 20  channel_published_at          24086 non-null  datetime64[ns, UTC]
 21  channel_country               24086 non-null  object
 22  channel_view_count            24086 non-null  float64
 23  channel_subscriber_count      24086 non-null  float64
 24  channel_have_hidden_subscribers 24086 non-null  int64
 25  channel_video_count           24086 non-null  float64
 26  channel_localized_title       24086 non-null  object
 27  channel_localized_description 24086 non-null  object
 28  reach_level                   24086 non-null  int64
 29  publish_hour                  24086 non-null  int32
 30  publish_day_of_week           24086 non-null  int32
 31  title_length                  24086 non-null  int64
 32  tag_count                     24086 non-null  int64
 33  duration_seconds              24086 non-null  float64
 34  is_shorts                     24086 non-null  int64
 35  channel_age_days              24086 non-null  int64
 36  channel_country_int           24086 non-null  int8
```

C. Final XGBoost model

```
[0] validation_0-mlogloss:1.06974   validation_1-mlogloss:1.07168
[100]   validation_0-mlogloss:0.30542   validation_1-mlogloss:0.41072
[200]   validation_0-mlogloss:0.19882   validation_1-mlogloss:0.34628
[300]   validation_0-mlogloss:0.15798   validation_1-mlogloss:0.32790
[400]   validation_0-mlogloss:0.13520   validation_1-mlogloss:0.32053
[500]   validation_0-mlogloss:0.11124   validation_1-mlogloss:0.31316
[600]   validation_0-mlogloss:0.09236   validation_1-mlogloss:0.30811
[700]   validation_0-mlogloss:0.07724   validation_1-mlogloss:0.30630
[721]   validation_0-mlogloss:0.07482   validation_1-mlogloss:0.30630
Report：
              precision    recall  f1-score   support

           1       0.96      0.92      0.94      3165
           2       0.48      0.63      0.55       386
           3       0.73      0.72      0.72       303

    accuracy                           0.88      3854
   macro avg       0.72      0.76      0.74      3854
weighted avg       0.89      0.88      0.89      3854
```

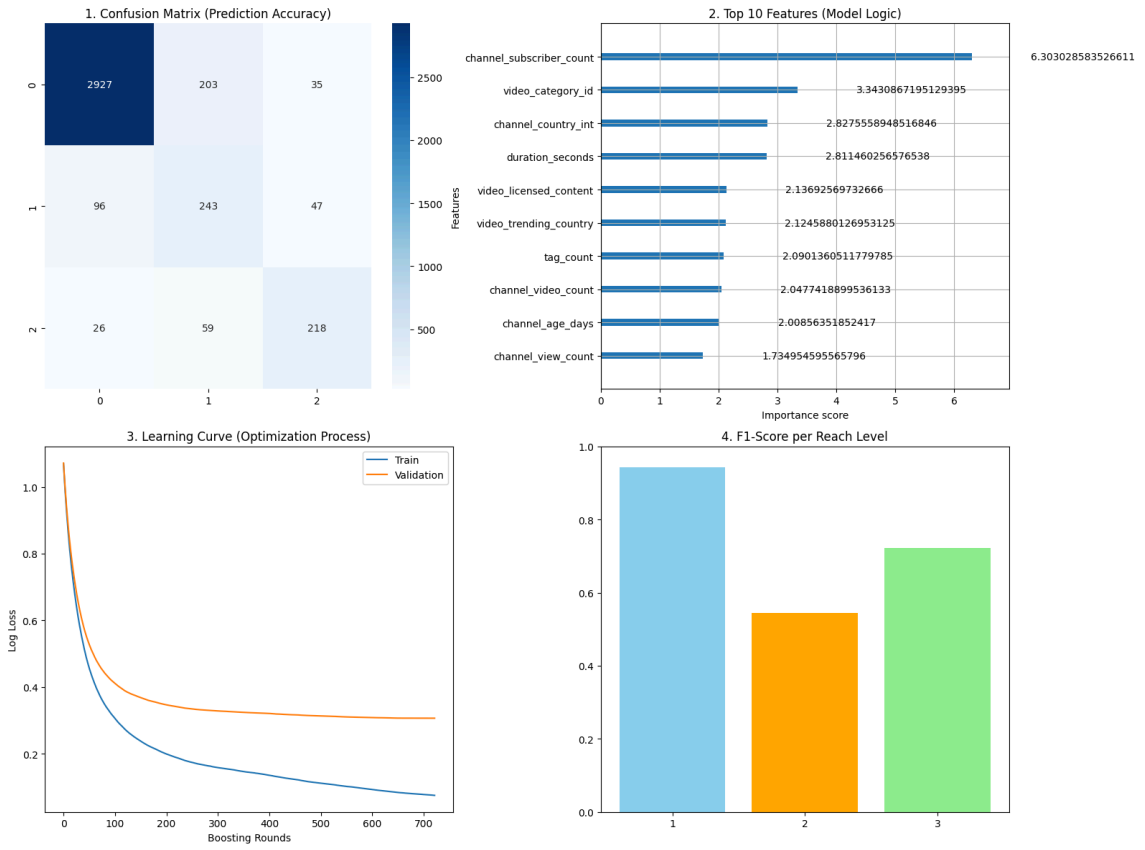
```python
model = xgb.XGBClassifier(
        n_estimators= 1000,
        learning_rate = 0.03,
        max_depth = 8,
        min_child_weight=0.1,
        early_stopping_rounds = 50,
        tree_method = 'auto',
        device = 'cpu',
        random_state = 42
)
```

D. Aggressive set of hyperparameters XGBoost model

```
[0] validation_0-mlogloss:1.07660   validation_1-mlogloss:1.07898
[100]   validation_0-mlogloss:0.29720   validation_1-mlogloss:0.42769
[200]   validation_0-mlogloss:0.14873   validation_1-mlogloss:0.33664
[300]   validation_0-mlogloss:0.09986   validation_1-mlogloss:0.31441
[400]   validation_0-mlogloss:0.07927   validation_1-mlogloss:0.30860
[500]   validation_0-mlogloss:0.06446   validation_1-mlogloss:0.30674
[600]   validation_0-mlogloss:0.05152   validation_1-mlogloss:0.30582
[640]   validation_0-mlogloss:0.04688   validation_1-mlogloss:0.30629
Report：
              precision    recall  f1-score   support

           1       0.96      0.93      0.95      3165
           2       0.49      0.60      0.54       386
           3       0.73      0.70      0.72       303

    accuracy                           0.88      3854
   macro avg       0.73      0.75      0.74      3854
weighted avg       0.89      0.88      0.89      3854
```

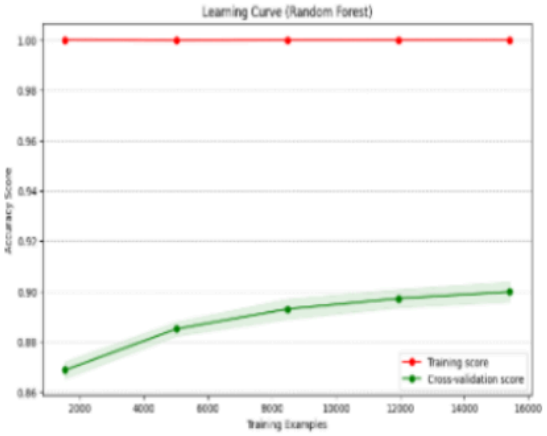
```python
model = xgb.XGBClassifier(
        n_estimators= 1500,
        learning_rate = 0.02,
        max_depth = 10,
        min_child_weight=0.1,
        early_stopping_rounds = 50,
        tree_method = 'auto',
        device = 'cpu',
        random_state = 42
)
```

E. Visualization of XGBoost

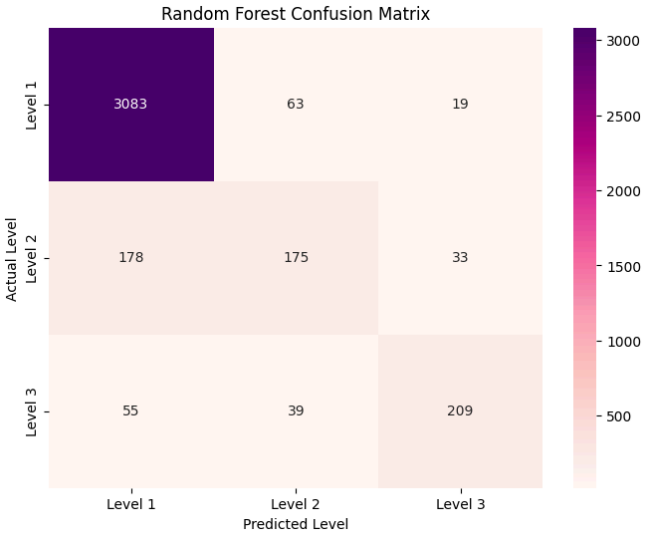


F. Random Forest Classification Report & learning Curve

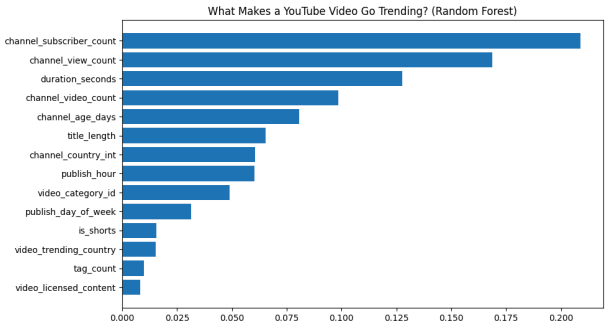
```
--- Random Forest Classification Report ---
              precision    recall  f1-score   support

           0       0.93      0.97      0.95      3165
           1       0.63      0.45      0.53       386
           2       0.80      0.69      0.74       303

    accuracy                           0.90      3854
   macro avg       0.79      0.71      0.74      3854
weighted avg       0.89      0.90      0.89      3854
```



G.Confusion Matrix



H. **Feature Importance**


What Makes a YouTube Video Go Trending? (Random Forest)
channel_subscriber_count
channel_view_count
duration_seconds
channel_video_count
channel_age_days
title_length
channel_country_int
publish_hour
video_category_id
publish_day_of_week
is_shorts
video_trending_country
tag_count
video_licensed_content
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0.200